A060 v0.2.1
Cross-Paper Vortex Mechanics and Stability
Frozen Analytic Tranche M00–M04

SST-Workbench – scientific protocol

13 October 2026

1 Identification and authority

This is an official SST Falsifier Framework v1.0.7 instance. Mathematical identities are tested as *analytic controls*, not independent evidence of SST particle physics or gravity. Historic A060 v0.1.0, A045 and A054 remain untouched.

2 Research question

Can dimensionless Euler-equilibrium, shear/pressure, projected-vorticity-amplification and local scale-flux identities be reproduced within frozen tolerances and with real native C++ FP64 parity?

3 Hypotheses

H_0: one or more analytic identities or backend parity conditions fails at fixed tolerance. H_1: the eligible reference identities and native parity tests agree with predictions. Neither hypothesis asserts nucleon stability, physical gravity, or vortex self-confinement.

4 Assumptions

Smooth incompressible velocity fields; toroidal analytic region for Landreman; 2π-periodic Fourier box for VA projection; linear velocity fields for exact spherical flux quadrature. The four PDFs are SHA-locked methodological sources, not computational datasets. Distinct papers do not imply independent numerical replication.

5 Symbols and dimensions

$A_{ij} = \partial_j u_i$ $[T^{-1}]$, $S = (A + A^T)/2$ $[T^{-1}]$, $\omega = \nabla \times u$ $[T^{-1}]$, Q_A $[T^{-2}]$, R_s, V_s $[T^{-3}]$ and Φ_ℓ $[L^2T^{-3}]$. Code uses nondimensional reference fields.

6 Frozen mathematical equations

$$\nabla \cdot u = 0, \qquad (u \cdot \nabla)u = -\nabla Q_L, \qquad Q_L = (x^2 + y^2 + 4z^2)/2, \tag{1}$$

$$Q_A = -\tfrac{1}{2}\mathrm{tr}(A^2), \qquad 2Q_A = \tfrac{1}{2}|\omega|^2 - S : S, \tag{2}$$

$$f = 1 - \sum_i |\lambda_i|^2/\|A\|_F^2, \qquad m = \max(2|Q_A|/\|A\|_F^2, (2|R_A|/\|A\|_F^3)^{2/3}), \tag{3}$$

$$m \le 1 - f \le 3m, \qquad \mathrm{VA} = \frac{\omega \cdot S\omega}{|\omega|^2}\omega, \tag{4}$$

$$\mathbb{P}_\perp \mathrm{VA} = \mathrm{VA} - \nabla\nabla^{-2}\nabla \cdot \mathrm{VA}, \tag{5}$$

$$\Phi_\ell = \frac{\ell^2}{10}\left[3R_s + \frac{1}{4}V_s\right], \qquad R_s = -\frac{1}{3}\mathrm{tr}(S^3), \qquad V_s = \omega \cdot S\omega. \tag{6}$$

The last relation is checked for globally linear fields, for which the omitted higher-order angular remainder is zero.

Machine-readable science contract

Research question Do independent dimensionless analytic reference models satisfy the M00–M04 Euler, pressure, strain, VA-projection and spherical-flux identities within predetermined tolerances?

Objective Qualify numerical diagnostic operators using four SHA-locked papers and analytical controls; do not infer SST particle/gravity validity.

H_0 H0: at least one analytic identity, source hash, or native FP64 parity condition fails at its preregistered threshold.

H_1 H1: all eligible M00–M04 local mathematical benchmarks converge and strict CPU FP64 parity matches the Python reference.

Registered equations

E1: Landreman Euler residual

$$\nabla \cdot u = 0, \quad (u \cdot \nabla)u = -\nabla Q_L, \quad Q_L = (x^2 + y^2 + 4z^2)/2 \tag{7}$$

Dimensional check: nondimensional reference coordinates/velocities

E2: Pressure/Schur tensor invariants

$$Q_A = -\tfrac{1}{2}\mathrm{tr}(A^2), \quad 2Q_A = \tfrac{1}{2}|\omega|^2 - S : S, \quad m \le 1 - f \le 3m \tag{8}$$

Dimensional check: Q_A and omega^2 both T^-2; f,m dimensionless

E3: Projected VA source

$$\mathbb{P}_\perp VA = VA - \nabla\nabla^{-2}\nabla \cdot VA, \quad \nabla \times g = -\mathbb{P}_\perp VA \tag{9}$$

Dimensional check: vorticity source T^-2; spatial operators cancel appropriately

E4: Zhang-Yao angular flux for linear fields

$$\Phi_\ell = \ell^2(3R_s + V_s/4)/10, \quad R_s = -\mathrm{tr}(S^3)/3, \; V_s = \omega \cdot S\omega \tag{10}$$

Dimensional check: ell^2 times T^-3 gives L^2 T^-3

E5: Strict FP64 cross-backend parity

$$e_\infty = \max_i |d_i^{Cpp} - d_i^{Py}| / \max(1, |d_i^{Py}|) \le 10^{-10} \tag{11}$$

Dimensional check: dimensionless normalized relative error

Registered code-to-mathematics steps

1. **S0** [G1; equations E1]: Verify four included PDF SHA-256 and original framework package-manifest files
2. **S1** [G2; equations E1]: Evaluate three Landreman configurations by central-difference Euler residuals
3. **S2** [G2; equations E2]: Evaluate shear/balanced vortex and preregistered random tensor wedge
4. **S3** [G3; equations E3, E4]: Evaluate periodic VA projection and direct sphere quadrature
5. **S4** [G5; equations E5]: Compile dedicated double C++ tensor diagnostics and compare against Python without fallback

7 Algorithm

Verify source hashes and framework manifest; test Landreman at three anisotropy values; test shear/balanced controls and 2,500 preregistered random matrices; verify the Fourier VA projection; compare spherical quadrature against invariant prediction; compile independent C++17 double kernel and check 29 tensor controls. All eligible analytic submodules run irrespective of earlier failures.

8 Primary sources and provenance

Chiarini `https://arxiv.org/abs/2609.39826`; Massey `https://arxiv.org/abs/2609.32543`; Landreman `https://arxiv.org/abs/2609.26742`; Zhang and Yao `https://arxiv.org/abs/2610.11029`. Each exact PDF is bundled under `sources/papers`; hashes appear in `source_contract.json` and the locked reference manifest. A045 registry identity conflict remains unresolved.

9 Blind gates

Gate	Scientific interpretation
G0	Framework frozen protocol integrity.
G1	M00 paper hash and embedded framework 92/92 SHA-256.
G2	M01 exact Euler and M02 pressure/shear tensor identities.
G3	M03 VA projection and M04 analytic spherical-flux identity.
G4/G7	Held-out and physical cross-provider replication disabled in this analytic tranche.
G5	Required independent native C++ double precision parity, no fallback.
G6	GPU screening disabled; no surrogate FP64 claim.
G8	Physical PKLSA state selection and A045 registry reconciliation: unresolved.
G9	Reveal deferred; no target identities imported.

10 Numerical thresholds

M01: $|\nabla \cdot u| < 3 \times 10^{-7}$, $|R_E| < 2 \times 10^{-7}$, pressure $< 3 \times 10^{-6}$; M02: wedge $< 10^{-10}$, balanced pressure $< 10^{-12}$; M03: curl projection $< 10^{-8}$ and mean $< 10^{-9}$; M04: relative quadrature $< 10^{-12}$; C++ parity $\le 10^{-10}$. Seeds 20261013 and 7189 are fixed.

11 Backend evidence

Python NumPy FP64 is reference only; C++17 pybind11 compiled using the user's toolchain is strict certification for the tensor kernel. OpenMP or serial actual backend must be reported. Unavailable toolchain yields UNRESOLVED rather than Python fallback. SYCL is not part of this tranche.

12 Blindness and conflicts

The pre-reveal scoring contains no canonical SST particle masses, preferred knot winners or gravity targets. Four methodological PDF files are public inputs. The historical A045 family identifier conflict is visible rather than overwritten; A054 state selection is not copied into the score.

13 Uncertainty and nulls

Analytical tolerances are absolute/rescaled FP64 bounds, not p-values. Synthetic random matrices are implementation controls. No independent turbulence replication is claimed.

14 Automatic results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	840c6959253a11e4868cc4d860fc5700657e95dfe4d633afe0289a5872590e31
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	365abf8b86bb26b3b579ae8ff8c70f97f1c88d0626e54c57cc1d882847c40178

Gate ledger

Gate	Status	Question
G0	PASS	Is the official v1.0.7 frozen protocol intact?
G1	PASS	Are four papers and framework hashes valid (M00 sources; registry warning separate)?
G2	PASS	Do Landreman Euler and shear/pressure tensor analytic benchmarks pass (M01,M02)?
G3	PASS	Do VA projection and scale-energy transfer analytic benchmarks pass (M03,M04)?
G4	NOT_APPLICABLE	Is independent held-out physical corroboration supplied?
G6	NOT_APPLICABLE	Optional GPU screening
G7	NOT_APPLICABLE	Independent source/provider replication outside analytic controls
G5	PASS	Does real compiled C++ double-precision tensor backend pass numerical parity?
G8	UNRESOLVED	Is physical PKLSA vortex dynamics and A045/A054 identity reconciled?
G9	DEFERRED	Reveal only after frozen gate review

Backend manifest


```
{
  "cpp": {
    "actual_backend": "openmp",
    "certified": true,
    "compile_fingerprint": "2abf520c79a01eb76e8df7301ad77f56e965257e016bc7a79aa8549806ccf823",
    "max_relative_parity": 2.6645352591003757e-15,
    "n_cases": 29,
    "precision": "float64",
    "requested_backend": "cpp"
  },
  "framework": "1.0.7",
  "modes": "M00-M04_ANALYTIC_ONLY",
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
```

```
  "schema": "SST-BACKEND-MANIFEST-2"
}
```


Source manifest


```
{
  "manifest_sha256": "399bb3c4268f2e63868e51788cff4a3725f998d95f66a014dbc52b918fe1756f",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A060",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "P3_Chiarini",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A060_Dynamical_Topolo
      "provenance_family": "paper_2609.39826v1",
      "registry_hint": {
        "catalog_id": "A060",
        "hits": []
      },
      "role": "published_method_and_equation_reference",
      "sha256": "a9e405021745c0f480fdfe27bc5ba0000022ad1c4793a2dc4bd17a09f733f339",
      "source_id": "P3_Chiarini",
      "version": "2609.39826v1"
    },
    {
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      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "P4_Massey",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A060_Dynamical_Topolo
      "provenance_family": "paper_2609.32543v1",
      "registry_hint": {
        "catalog_id": "A060",
        "hits": []
      },
      "role": "published_method_and_equation_reference",
      "sha256": "995fefe072800ce0d56cf0970515bceef1c505ac9b0da5334fa304c28bac309c",
      "source_id": "P4_Massey",
      "version": "2609.32543v1"
    },
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      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "P7_Landreman",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A060_Dynamical_Topolo
      "provenance_family": "paper_2609.26742v2",
      "registry_hint": {
        "catalog_id": "A060",
        "hits": []
      },
      "role": "published_method_and_equation_reference",
      "sha256": "3465595b0aa0e954e1885f94c898b6cb76c87220e446b5c1b11b087cef0f18d4",
      "source_id": "P7_Landreman",
      "version": "2609.26742v2"
    },
    {
      "catalog_id": "A060",
      "evidence_class": "independent_source",
```

```
      "exists": true,
      "independence_family": "P9_ZhangYao",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A060_Dynamical_Topolo
      "provenance_family": "paper_2610.11029v1",
      "registry_hint": {
        "catalog_id": "A060",
        "hits": []
      },
      "role": "published_method_and_equation_reference",
      "sha256": "133b64d4487e603b1a7ecc3cee5f179e8542f6cf5e3caa048f59ff101cba0223",
      "source_id": "P9_ZhangYao",
      "version": "2610.11029v1"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation and limitations

Reference PASS validates a local mathematical implementation only. Physical dynamical stability, free-space self-confinement and gravity derivations remain outside the tested domain. A failing analytic submodule does not silently suppress the other independent submodules.

16 Reproducibility

Use `run_all.cmd` on Windows. The bundled v1.0.7 framework is SHA-verified against the Google Drive package manifest. `FROZEN_PROTOCOL.json`, BLIND output, report TeX and archive hashes supply immutable provenance. Re-running on Windows rebuilds the native C++ extension from source.